

OLED

Voltage input must be 3.3V

Boost

Voltage input must be 3.3V

Connectors

Voltage and signals input must be 3.3V

Options

By default CS is held low for I2C.
Cut allow MCU control of CS for 4WSPI mode

Slave Address (7bit) is set according to:
D/C : 7 bit address (HEX)

0 : 0x3C (Default)
1 : 0x3D

The BS1 pin determines I2C (HIGH, Default)
or 4WSPI (LOW) interface selection.

For I2C mode this jumper needs
to be closed. Open for 4WSPI mode

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KiCad E.D.A. eeschema 7.0.6

Rev: Id: 1/1

OLED

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Note: DS called for 910k ohms here

Boost

$CC = 1.25 \cdot (1 + R_1 / R_2) \cdot (1.25 \cdot (1 + 47/5.6)) / 11.74\text{ V}$

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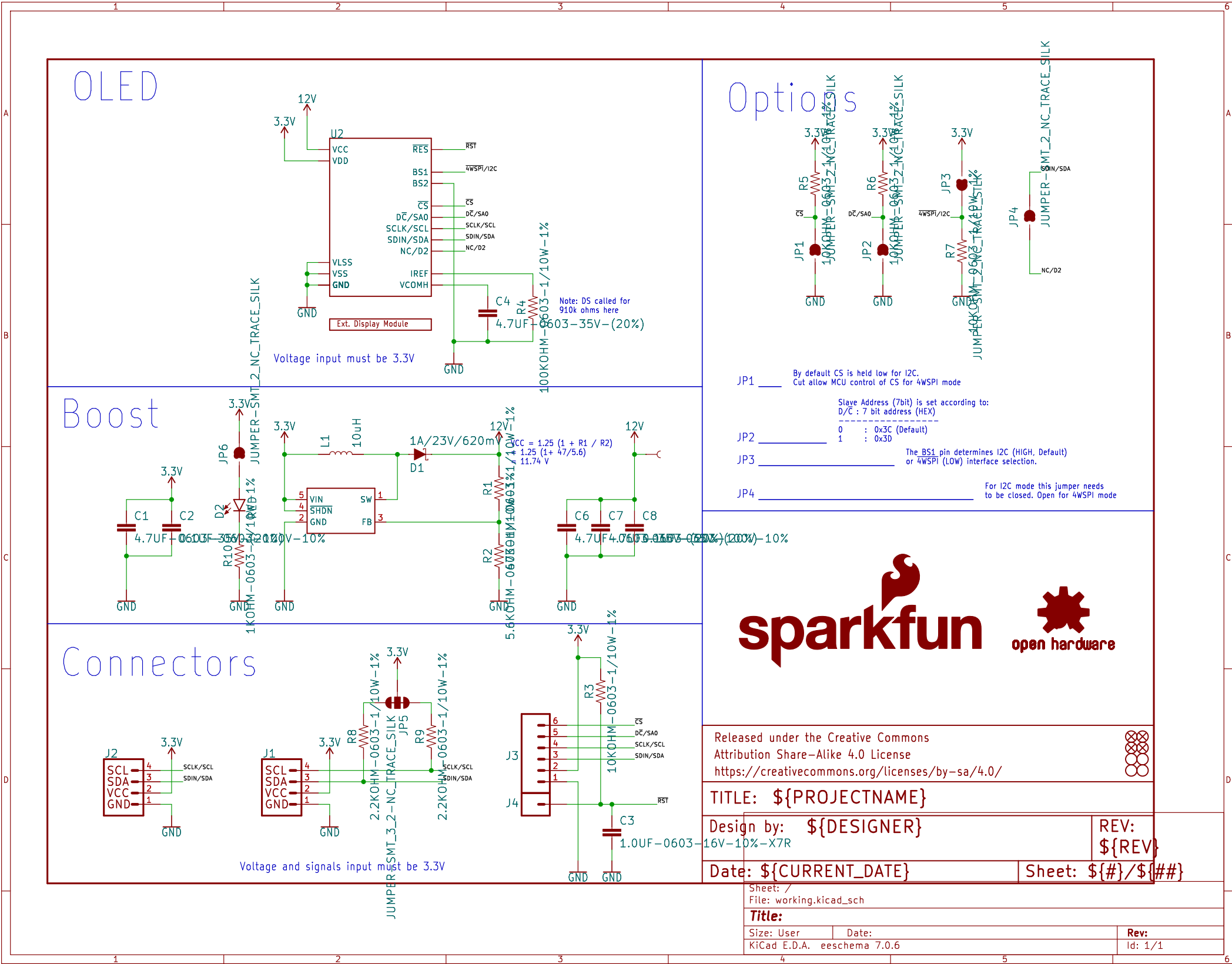
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The image shows a schematic diagram for a SparkFun board, divided into four main sections: OLED, Boost, Connectors, and Options.

OLED: This section shows the connection of an external display module (U2) to the board's I2C bus. The module's VCC and GND are connected to the board's 3.3V and GND pins. Its SDA and SCL pins are connected to the board's SDA and SCL pins via jumpers JP1 and JP2. A pull-up resistor R4 (10KOHM-0603-1%/10W-1%) is connected between the SDA/SCL line and the 3.3V supply. A note indicates that the DS component is called for 910k ohms here.

Boost: This section shows a boost converter circuit (D1, L1, SW, FB) that takes a 3.3V input and outputs 12V. It includes several capacitors (C1, C2, C3, C4, C6, C7, C8) and resistors (R1, R2, R3). A note specifies that the output voltage is 12V ± 1%.

Connectors: This section shows the connections for J1, J2, J3, and J4. J1 and J2 are headers for the 3.3V supply and ground. J3 and J4 are headers for the I2C bus (SDA, SCL) and the reset pin (RST).

Options: This section provides information about the jumper settings (JP1, JP2, JP3, JP4) and the slave address (7bit) set according to D/C : 7 bit address (HEX). It also mentions that the BS1 pin determines I2C (HIGH, Default) or 4WSPI (LOW) interface selection.

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[illegible]

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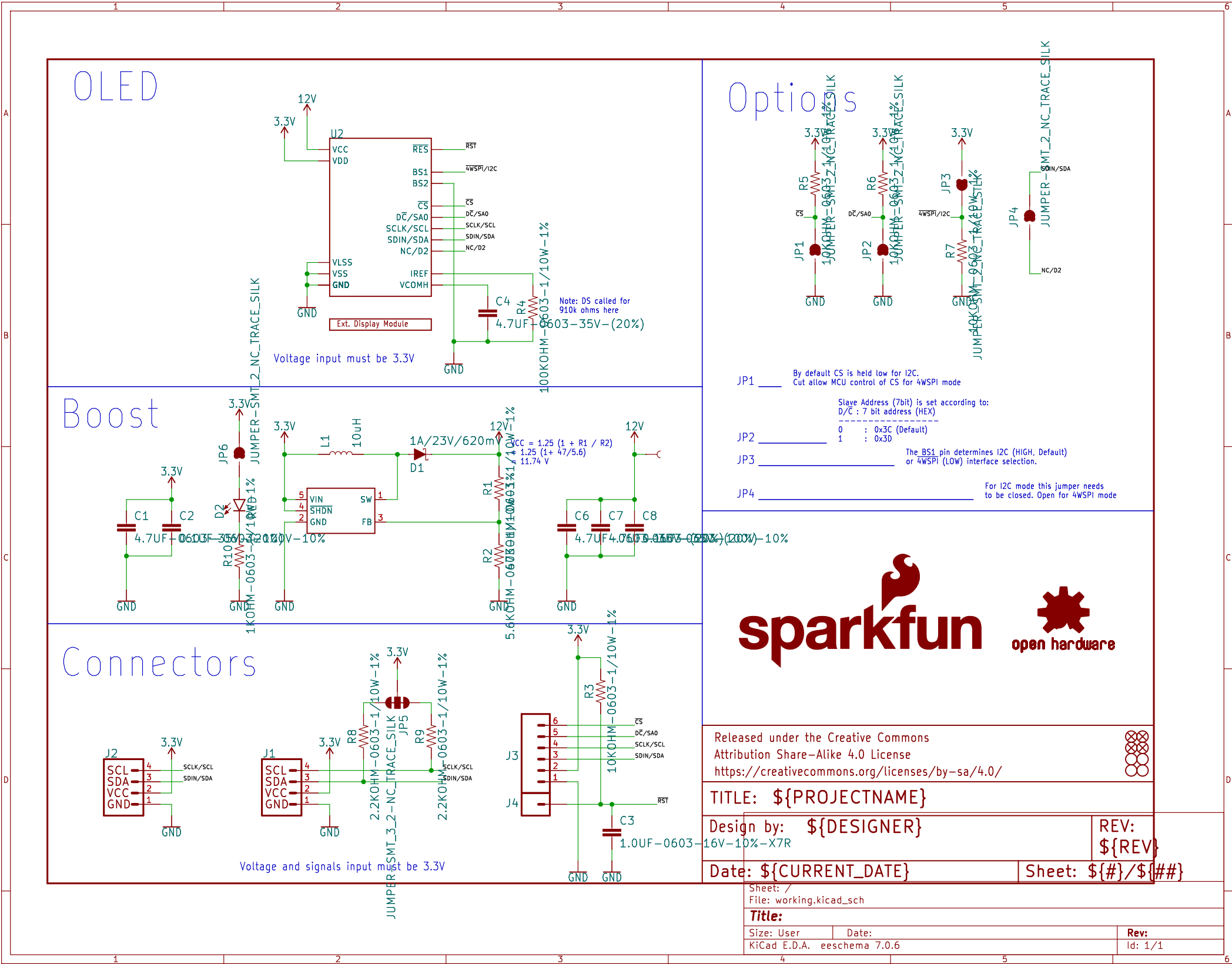
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12V

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